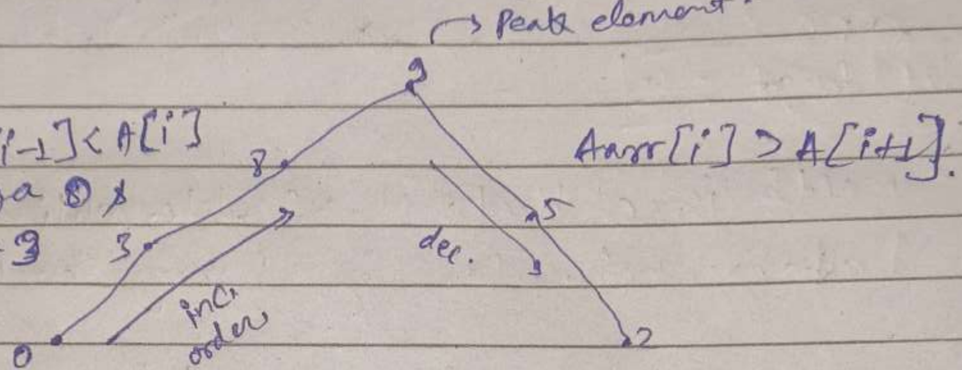
Q 6. Peak Index in mountain Array :-

arr = [0, 3, 8, 9, 5, 2]



maths, 9 bada hoga ~~8~~ 8 se
8 bada hoga 3 se ---

- Peak element jo hota hai wo leffe se bhi bada hai or left se bhi bada hai.

2. $\boxed{Arr[p-1] < A[p] > A[p+1]}$ → peak.

Note :- In a mountain array pehla element and last element kabhi bhi peak element nhi ho skta.

$\boxed{idx = 0, n-1 \neq peak}$.

To by Binary seach :-

0	3	8	9	5	2

↑ Start ↑ mid. ↑ end.

$$Mid = S + \frac{(e-s)}{2}.$$

Ab hme check karna hai ki 8 peak element hai ki nhi or peak element ki condition hme ~~pta~~ pata hai.

[A[mid-1] < A[mid] > arr[mid+1]]

→ return mid.

- Ab binary search ki help se pta karo ki peak element hamara left me hoga ki right me.

mid.

left (inc.)
(arr[mid-1] < A[mid].)
→ i.e right search.
toh. start = mid+1

right (dec.)
(arr[mid] < arr[mid+1].)
→ i.e left search.
toh. end = mid-1.

basic of binary search

A. Pseudocode:- start = 1 end = arr-2 // kiyoki first & last element peak element nhi ho skta.

```
while (st <= end) {
    mid = s + (e-s)/2
  if (arr[mid-1] < arr[mid] > arr[mid+1])
        return mid;
  if (arr[mid-1] < arr[mid]) // inc. order → right side
        st = mid+1.
  else // dec. order → left side.
        end = mid-1.
```